

Set-B

 BRAC University Department of Computer Science and Engineering Semester: Spring 2026 Course Code: CSE460 Course Title: VLSI Design	Midterm Examination Full Marks: 10 x 2 + 15 = 35 Time: 1 hour 20 minutes Date: 30 March 2026
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Student ID:	Name:	Section:
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[Answer all **THREE** questions. **Return** the question paper with the answer script.]

Question 1: [CO2]

Simon is required to design a sequence detector that can detect the patterns **101** and **110** from a serial data stream. When the sequence 101 is detected and the Red LED (R) turns on; when 110 is detected, the Blue LED (B) turns on.

You may use the variables **R** and **B** to represent the status of the corresponding LEDs.

A sample test is given below to show the detector circuit's functionality:

Clk	1	2	3	4	5	6	7	8	9	10	11	12
w	1	0	1	1	0	0	1	1	0	1	0	0
R	0	0	1	0	0	0	0	0	0	1	0	0
B	0	0	0	0	1	0	0	0	1	0	0	0

(a)	Determine the type of FSM that would be needed for the given problem and design the state diagram for the corresponding FSM. Using more states than necessary will result in a deduction of marks.	[5]
(b)	Derive the state-assigned table using the Gray encoding from your state diagram in (a).	[3]
(c)	Find the expression for the outputs (R and B) for the FSM using KMAP.	[2]

Note: Use the table below for Gray coding if needed.

Decimal Equivalent	Binary Code			Gray Code		
	B2	B1	B0	G2	G1	G0
0	0	0	0	0	0	0
1	0	0	1	0	0	1
2	0	1	0	0	1	1
3	0	1	1	0	1	0
4	1	0	0	1	1	0
5	1	0	1	1	1	1
6	1	1	0	1	0	1
7	1	1	1	1	0	0

Question 2: [CO2]

A digital communication system transmits data bits serially through a channel. To enable simple error detection at the receiver, a **parity bit** is generated for every group of three transmitted bits. A circuit receives a serial input w , which represents the incoming data bits. After observing **three consecutive bits**, the circuit generates an **even parity bit P** . The parity bit is defined as:

- $P=1$ if the number of **1s** in the three-bit sequence is **even**.
- $P=0$ otherwise

After producing the parity bit for three consecutive incoming bits, the circuit immediately starts processing the **next group of three incoming bits**. The table below shows a sample operation:

w	1	1	0	1	0	0	1	0	1	0	0	0	1	1	1	1
P	0	0	0	1	0	0	0	0	0	1	0	0	1	0	0	0

(a)	Design the state diagram for the corresponding FSM with a minimum number of states . How many Flip-Flops are required to implement the circuit? [Hint: To minimize the number of states, different input sequences that lead to the same combination (number of bits observed and parity so far) can share the same state.]	[6]
(b)	Derive the state-assigned table from the state diagram using Binary Encoding .	[3]
(c)	Find the expression for the output logic for the FSM using KMAP.	[1]

Question 3: [CO1]

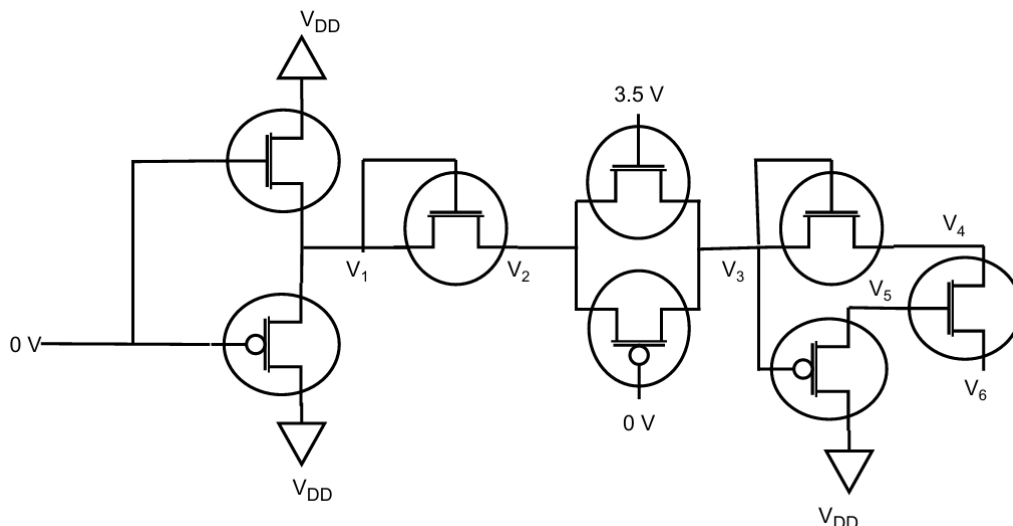


Figure 1

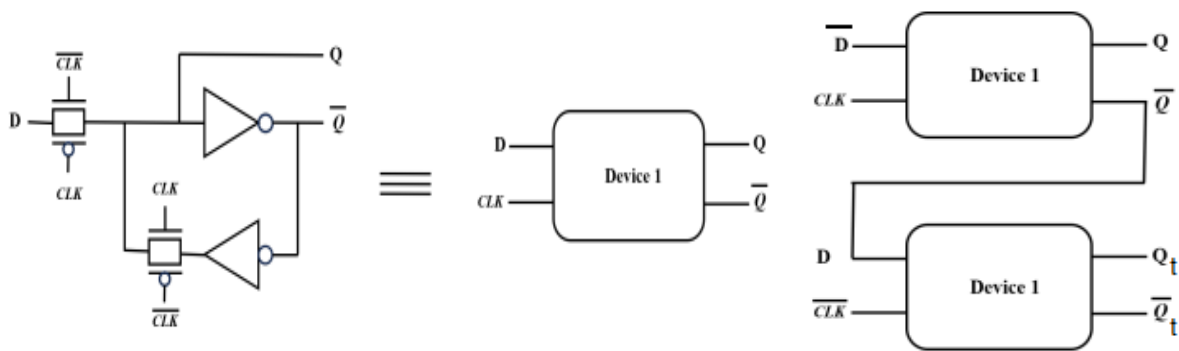
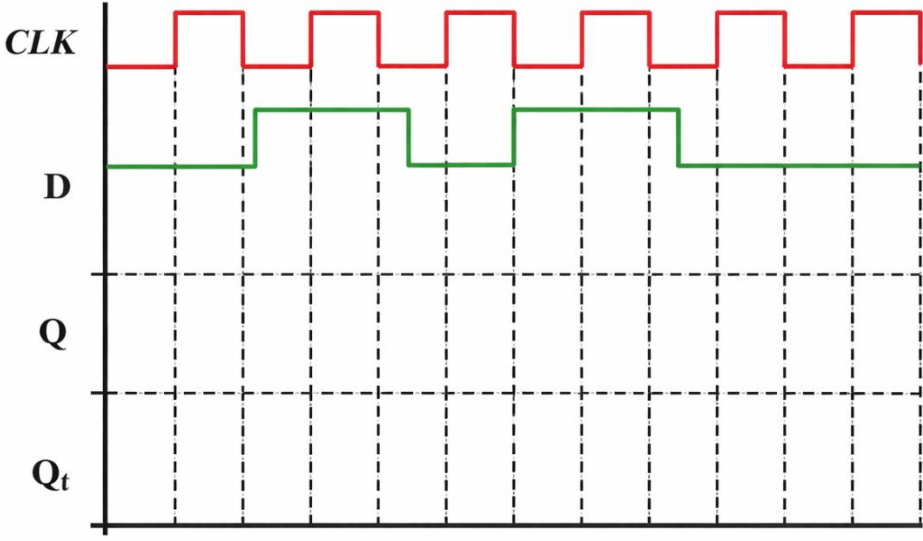


Figure 2

(a)	Suppose you are tasked with designing a 4:1 mux. The client requires the design of the mux to be such that it includes the lowest number of transistors possible and the circuit can pass both good '0' and good '1'. Design the mux and specify the number of transistors used. [Notes: All the input signals are also available in its complement form]	[5]
(b)	If $V_{DD} = 5V$, $V_{tn} = 0.5V$ and $V_{tp} = 0.6V$, find the node voltages from Figure 1.	[5]
(c)	For the circuit in Figure 2, draw the waveform of Q and Q_t and identify the circuit type. 	[5]